Part A- Introduction

I.	**Course Code and Title**	**: PHY 211 and Physics II**
II.	**Credit**	**: 3.0 Credit Hours**

1. Rationale of the Course:

To extend the student's knowledge and understanding of some of the laws, principles and theories of waves and oscillations, optics and modern physics. To develop an appreciation of how these laws, principles and theories operate in experiment and applied engineering problems.

2. Course Description:

Waves and Oscillations: Differential equation of simple harmonic oscillator and its solution, Total energy and average energy, Combination of simple harmonic oscillations, Spring mass system, Two body oscillation, Reduced mass, Damped oscillation, Forced oscillation, Resonance; Progressive wave, Power and intensity of wave, Stationary wave, Group velocity and phase velocity.

Optics: Theories of light, Interference of light, Condition for constructive and destructive interference, Young's double slits experiment, expression for the width of bright fringe or dark fringe, Fresnel bi-prism and its application, Interference in thin films, Newton's rings, Wavelength determination by Newton's rings experiment, Interferometer and its working principle; Diffraction: Diffraction by single slit, Diffraction by double slits, Resolving power of optical instruments, Diffraction grating and its uses; Polarization, Production and analysis of polarized light, Brewster's Law, Malus Law, Polarization by double refraction, Nicol prism, its construction & working procedure, Optical activity and specific rotation, Polarimeter and its construction.

Modern Physics: Galilean relativity and Einstein's special theory of relativity; Lorentz transformation equations, Length contraction, Time dilation and mass-energy relation. Photoelectric effect, Compton effect; de Broglie matter waves and its success in explaining Bohr's theory, Pauli's exclusion principle, Constituent of atomic nucleus, Nuclear binding energy, Different types of radioactivity, Radioactive decay Law; Nuclear reactions, Nuclear fission, Nuclear fusion, Nuclear power plant.

Objective:

Having successfully completed this course, students' will be able to demonstrate knowledge and understanding of:

a) The basic laws that underlie the properties of optics.
b) Introduce the basic knowledge of waves and oscillations.
c) To know the basic concepts of modern physics.

3. Course Outcome (CO) Matrix:

Course Outcomes (CO) of the Course			
Course Outcome	**Bloom's Taxonomy**	**Domain**	**Program Outcome**
CO1: **Explain** the fundamentals of engineering principles related to optics, waves and oscillations, and modern physics.	Level -2 (Understand)	C	PO 1: Engineering Knowledge K1 (Natural Sciences)
CO2: **Applying** the basic knowledge of optics, waves and oscillations, and modern physics to demonstrate different physics laws.	Level -3 (Apply)	C	PO 2: Problem analysis K1 (Natural Sciences)

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork (These domains are cognitive (thinking), affective (emotion/feeling), and psychomotor (physical/kinesthetic)**)**

K1: knowledge profile, P2: complex engineering problem

4. Mapping of COs and Program Outcomes (POs):

COs	Program Outcomes H = High, M = Medium and W = Weak											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	H											
CO2		M										

5. Justification for CO-PO Mapping:

Mapping	Level	Justification
CO1-PO1	High	Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
CO2-PO2	Medium	Identify, formulate, research the literature and analyze complex engineering problems and reach substantiated conclusions using the principles of mathematics, the natural sciences and the engineering sciences.

6. Teaching Learning Strategy:

Teaching and Learning Activities	Engagement (Hours)
Face-to-Face Learning ➢ Lecture ➢ Practical / Tutorial / Studio ➢ Student-Centered Learning	42
Self-Directed Learning	
➢ Non-face-to-face learning	42
➢ Revision	21
➢ Assessment Preparations	21
Formal Assessment	
➢ Continuous Assessment	2.5
➢ Mid Term Exam ➢ Final Examination	1.5 3.0
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7. Teaching Methodology:

Lecture and Discussion, Co-operative and Collaborative Method, Problem Based Method

<u>Part B- Content of the Course</u>

8. Teaching Learning and Assessment Strategy Mapped with COs:

Class		Intended Learning Outcomes	Teaching/ Learning Strategy	Assessment Strategy	CLO's
Week 1	1	Brief discussion on Course Details	Discussion		1,2
	2	Motion, Waves and Oscillation	Lecture, Q/A, Feedback	Q/A, Assignment	
	3	Differential equation of simple harmonic motion (SHM) and its solution	Lecture, Q/A, Feedback	Q/A, Assignment	
2	4	Potential, Kinetic, and Total Energy of Oscillation	Lecture, Q/A, Feedback	Q/A, Assignment	
	5	Average Energy of a body executing SHM	Lecture, Q/A, Feedback	Q/A, Assignment	
	6	Composition of simple harmonic oscillation	Lecture, Q/A, Feedback	Q/A, Assignment	
3	7	Progressive wave, Stationery Wave, & Power and intensity of wave	Lecture, Q/A, Feedback	Q/A, Assignment	

	8	Spring mass system and Torsional Pendulum	Lecture, Q/A, Feedback	Q/A, Assignment	
	9	Two body oscillation and Reduced mass & **Class Test- 01**	Lecture, Q/A, Feedback	Q/A, Assignment	
4	10	Damped oscillation and Resonance	Lecture, Q/A, Feedback	Q/A, Assignment	
	11	Stationary Wave, Phase velocity, Group velocity	Lecture, Q/A, Feedback	Q/A, Assignment	
	12	Review of Waves and Oscillations & **Assignment**	Lecture, Q/A, Feedback	Q/A, Assignment	
5	13	Theories of light & **Class Test- 02**	Lecture, Q/A, Feedback	Q/A	
	14	Interference of light	Lecture, Q/A, Feedback	Q/A	
	15	Young's double slit experiment	Lecture, Q/A, Feedback	Q/A	
6	16	Fresnel bi-prism & Interference in thin films	Lecture, Q/A, Feedback	Q/A	
	17	Newton’s rings	Lecture, Q/A, Feedback	Q/A	
	18	Interferometers	Lecture, Q/A, Feedback	Q/A	
7	19	Diffraction, Diffraction by single slit	Lecture, Q/A, Feedback	Q/A	
	20	Diffraction by circular aperture and double slits	Lecture, Q/A, Feedback	Q/A	
	21	Diffraction grating, Resolving power of optical instruments	Lecture, Q/A, Feedback	Q/A	
Mid term					
8	22	Polarization: Production of polarized light, Brewster's Law	Lecture, Q/A, Feedback	Q/A	
	23	Malus Law, Polarization by double refraction	Lecture, Q/A, Feedback	Q/A	

	24	Nicol prism, Optical activity, Laser, Polarimeters	Lecture, Q/A, Feedback	Q/A	
9	25	Optical activity **& Class Test-03**	Lecture, Q/A, Feedback	Q/A	
	26	General discussion on relativity	Lecture, Q/A, Feedback	Q/A	
	27	Einstein's special theory of relativity	Lecture, Q/A, Feedback	Q/A	
10	28	Galilean transformation and Lorentz transformation equations	Lecture, Q/A, Feedback	Q/A	
	29	Time dilation (Relativity of time)	Lecture, Q/A, Feedback	Q/A	
	30	Length Contraction (Relativity of length)	Lecture, Q/A, Feedback	Q/A	
11	31	Einstein's Mass-Energy Relation	Lecture, Q/A, Feedback	Q/A	
	32	Photoelectric effect	Lecture, Q/A, Feedback	Q/A	
	33	Compton effect	Lecture, Q/A, Feedback	Q/A	
12	34	De Broglie matter wave	Lecture, Q/A, Feedback	Q/A	
	35	Pauli’s exclusion principle, Constituent of atomic nucleus and Nuclear binding energy	Lecture, Q/A, Feedback	Q/A	
	36	Radioactivity and its types	Lecture, Q/A, Feedback	Q/A	
13	37	Radioactive decay Law	Lecture, Q/A, Feedback	Q/A	
	38	Nuclear reactions, Nuclear fission and Nuclear fusion	Lecture, Q/A, Feedback	Q/A	
	39	Solving Mathematical Problems (Modern Physics)	Lecture, Q/A, Feedback	Q/A	

14	40	Nuclear power plant & **Class Test- 04**	Lecture, Q/A, Feedback	Q/A	
	41	Review Class of optics	Lecture, feedback		
	42	Review of Modern Physics	Lecture, feedback		

Part C- Assessment Strategy

Components		Grading	CO	Bloom's Taxonomy
Continuous Assessment (20%)	Test 01-03	10%	CO1, CO2	C2, C3
	Assignment & Class Participation	10%	CO1, CO2	C2, C3
Mid-term		30%	CO1, CO2	C2, C3
Final Examination		50%	CO1, CO2	C2, C3
Total Marks		100%		
(CO = Course Outcome, C = Cognitive Domain, P = Psychomotor Domain, A = Affective Domain)				

Part D- Learning Materials:

Text and Reference Books:

1. Physics for Engineers (Part I & II) - Dr. Giasuddin Ahmed
2. B.Sc. Physics (Volume I & II) - C.L. Arora
3. A Text Book of Optics - Brijlal and Subrahmanyam
4. Waves and Oscillation - Brijlal and Subrahmanyam
5. Concepts of Modern Physics – Arthur Beiser